

Practical No. 8

Aim: To create and configure a Virtual Private Cloud (VPC) in AWS and set up subnets, route tables, and internet gateway to manage network resources securely.

Theory:

Amazon Virtual Private Cloud (VPC) is a logically isolated section of the AWS cloud where you can launch AWS resources in a virtual network that you define. It gives you full control over the network configuration including IP address ranges, subnets, route tables, and gateways.

A VPC typically consists of public subnets (accessible from the internet via an Internet Gateway) and private subnets (accessible only within the VPC, using a NAT Gateway for outbound internet access). Route tables control how traffic flows between subnets and the internet.

Key Components of a VPC

- Subnet — A range of IP addresses within a VPC, either public or private
- Internet Gateway — Allows communication between the VPC and the internet
- NAT Gateway — Allows private subnet resources to access the internet without being publicly exposed
- Route Table — Contains rules that determine where network traffic is directed
- Security Group — Acts as a virtual firewall to control inbound and outbound traffic

Steps:

Task 1: Create the VPC

1. In the AWS Console, search for and open VPC. From the VPC Dashboard, click Create VPC and select VPC and more.
2. Set the name tag to lab, IPv4 CIDR to 10.0.0.0/16, 1 Availability Zone, 1 public subnet (10.0.0.0/24), 1 private subnet (10.0.1.0/24), NAT Gateway in 1 AZ, and VPC endpoints to None.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
lab

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 2 3

Customize AZs

Number of public subnets [Info](#)
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 1

Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 1 2

Customize subnets CIDR blocks

Public subnet CIDR block in us-east-1a
10.0.0.0/24 256 IPs

Private subnet CIDR block in us-east-1a
10.0.1.0/24 256 IPs

NAT gateways (\$) [Info](#)
Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway.

In 1 AZ 1 per AZ

VPC endpoints [Info](#)
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None S3 Gateway

3. Click Create VPC and wait for all resources to be created, then click View VPC to confirm the VPC, subnets, route tables, Internet Gateway, and NAT Gateway are all provisioned.

Task 2: Create Additional Subnets

1. In the left pane, choose Subnets and click Create subnet. Select lab-vpc, set name to lab-subnet-public2, choose a second Availability Zone (us-east-1b), and set CIDR to 10.0.2.0/24. Click Create subnet.
2. Create another subnet named lab-subnet-private2 in the same Availability Zone with CIDR 10.0.3.0/24.

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

lab-subnet-public2
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1b)

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.2.0/24

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

lab-subnet-private2

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1b)

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.3.0/24

- In Route tables, select lab-rtb-private1-us-east-1a, go to Subnet associations, click Edit subnet associations, and also select lab-subnet-private2. Save associations.

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	lab-subnet-public2	subnet-06d369d33722fb44e	10.0.2.0/24	–	Main (rtb-0ed0480e3d2783089)
☒	lab-subnet-private1-us-east-1a	subnet-0080d7bd31881b4da	10.0.1.0/24	–	rtb-03989d4f8310791e9 / lab-rtb-private1-us-east-1a
☐	lab-subnet-public1-us-east-1a	subnet-00676d824d0ceb8d8	10.0.0.0/24	–	rtb-0f735ebff3dbdbfb4 / lab-rtb-public
☒	lab-subnet-private2	subnet-084d6fa7b12873aa2	10.0.3.0/24	–	Main (rtb-0ed0480e3d2783089)

- Select lab-rtb-public, go to Subnet associations, click Edit subnet associations, and also select lab-subnet-public2. Save associations.

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	lab-subnet-public2	subnet-06d369d33722fb44e	10.0.2.0/24	–	Main (rtb-0ed0480e3d2783089)
☐	lab-subnet-private1-us-east-1a	subnet-0080d7bd31881b4da	10.0.1.0/24	–	rtb-03989d4f8310791e9 / lab-rtb-private1-us-east-1a
☒	lab-subnet-public1-us-east-1a	subnet-00676d824d0ceb8d8	10.0.0.0/24	–	rtb-0f735ebff3dbdbfb4 / lab-rtb-public
☐	lab-subnet-private2	subnet-084d6fa7b12873aa2	10.0.3.0/24	–	rtb-03989d4f8310791e9 / lab-rtb-private1-us-east-1a

Task 3: Create a VPC Security Group

- In the left pane, choose Security groups and click Create security group.
- Set Security group name to Web Security Group, Description to Enable HTTP access, and select lab-vpc.

Basic details

Security group name [Info](#)

Web Security Group

Name cannot be edited after creation.

Description [Info](#)

Enable HTTP access

VPC [Info](#)

vpc-0a74f17a090aac752 (lab-vpc)

- Under Inbound rules, add a rule with Type: HTTP, Source: Anywhere-IPv4, and Description: Permit web requests. Click Create security group.

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional
HTTP	TCP	80	Anywh...	Permit web requests

Task 4: Launch a Web Server Instance

- Navigate to EC2 and click Launch instance. Name the instance Web Server 1, select Amazon Linux 2023 AMI, and keep instance type as t2.micro.

Name and tags Info

Name

Web Server 1

▼ **Application and OS Images**

An AMI contains the operating system, applications, and data. You can select an AMI from the catalog or choose [Browse more AMIs](#).

Search our full catalog including 1000s of AMIs

AMI from catalog Recents

Name

Amazon Linux 2023 kernel-6.1 AMI

▼ **Instance type** Info | Get advice

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation

On-Demand Linux base pricing: 0.0116 USD per hour

On-Demand Windows base pricing: 0.0162 USD per hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per hour

On-Demand SUSE base pricing: 0.0116 USD per hour

Additional costs apply for AMIs with pre-installed software

▼ **Key pair (login)** Info

You can use a key pair to securely connect to your instance.

Key pair name - *required*

vockey

- Under Network settings, select lab-vpc, choose lab-subnet-public2, enable Auto-assign public IP, and select Web Security Group.
- In Advanced details, paste the provided bootstrap script in the User data box to install and start the Apache web server automatically on launch.

User data - optional Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
# Install Apache Web Server and PHP
dnf install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```

▼ **Network settings** Info

VPC - *required* Info

vpc-0a74f17a090aac752 (lab-vpc)

Subnet Info

subnet-06d369d33722fb44e lab-subnet-public2

VPC: vpc-0a74f17a090aac752 Owner: 381492002005 Availability Zone: us-east-1b (use1-az4)

Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.2.0/24

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic.

Create security group Select existing security group

Common security groups Info

Select security groups

Web Security Group sg-06b19352f580564fb X

VPC: vpc-0a74f17a090aac752

- Click Launch instance. Once the instance shows 2/2 status checks passed, copy the Public IPv4 DNS and open it in a browser to verify the web server is running.

Not secure 54.87.230.152

aws Load Test RDS

Meta-Data	Value
InstanceId	i-0696645b3e177bcd
Availability Zone	us-east-1b

Current CPU Load: 1%

Conclusion: Thus, we have successfully created and configured a Virtual Private Cloud (VPC) in AWS with subnets, route tables, and internet gateway to manage network resources securely.